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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER Discipline Specific Core Course (DSC) - MG4DSCZGY203 - BIOLOGICAL BASIS OF BEHAVIOR – IV

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember (K), Understand (U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A Multiple Choice Questions Answer all questions Each question carries 1 marks

1. Which lymphocytes mature in the bone marrow?
Answer: b) B lymphocytes

2. Which immunoglobulin is mainly transferred through breast milk?
Answer: c) IgA

3. Explain Type IV hypersensitivity reaction?
Answer: a) Cell-mediated immunity

4. Androgens mainly influence which function?
Answer: b) Development of male secondary characters

5. Increased muscle mass in males during puberty is mainly due to:
Answer: d) Testosterone

6. Estrogen helps in regulation of:
Answer: b) Menstrual cycle

7. Increase in sex hormone secretion occurs mainly during:
Answer: b) Puberty

8. For someone who travels across time zones, what is the most immediate way to help reset their circadian rhythm?
Answer: b) Seeking exposure to daylight at the appropriate local time

9. Narcolepsy is a disorder marked by:
Answer: b) Sudden sleep attacks

10. Which brain structure is responsible for converting neural signals from the SCN into melatonin secretion?
Answer: c) Pineal gland

Part B Short Answer Questions Answer any 10 questions Each question carries 2 marks

11. Write briefly about the spleen as a secondary lymphoid organ.

The spleen is a major secondary lymphoid organ located in the upper left abdomen. It filters blood and removes old or damaged red blood cells. It also plays an important role in immune response by producing lymphocytes. The spleen traps blood-borne pathogens and helps in antibody production.

12. What is rheumatoid arthritis?
Rheumatoid arthritis is a chronic autoimmune disease that primarily affects joints. The immune system mistakenly attacks the synovial membrane, causing inflammation. This leads to pain, swelling, and stiffness in joints. Over time, it can result in joint deformity and reduced mobility.

13. Mention two stress-reducing practices that improve immunity.
Practices such as regular physical exercise and meditation help reduce stress. Exercise improves circulation and boosts immune cell function. Meditation and yoga reduce cortisol levels, which can suppress immunity. Both practices enhance overall physical and mental health.

14. Mention any two causes of immunodeficiency.

Immunodeficiency can be caused by genetic disorders affecting immune cells. It can also result from infections like HIV that damage the immune system. Malnutrition weakens immune responses. Certain medications such as chemotherapy drugs can also suppress immunity.

15. Write a short note on the scrotum.

The scrotum is a pouch of skin that contains the testes in males. It helps regulate temperature, keeping the testes cooler than body temperature. This temperature control is essential for sperm production. It also provides protection to the testes.

16. What is the excitement phase of the sexual response cycle?
The excitement phase is the initial stage of the sexual response cycle. It involves physical and psychological arousal. In males, it leads to erection, while in females, it causes lubrication and swelling of genital tissues. Heart rate and blood flow also increase during this phase.

17. State the functions of Bartholin's glands.
Bartholin's glands are located near the vaginal opening. They secrete mucus to lubricate the vagina. This lubrication helps during sexual intercourse. It also maintains moisture and reduces friction.

18. Describe the process of development of male reproductive organs during fetal life.
Male reproductive organs develop under the influence of testosterone during fetal life. The testes form in the abdomen and later descend into the scrotum. The Wolffian ducts develop into male reproductive structures like vas deferens. The presence of the Y chromosome initiates this process.

19. Describe any two symptoms of mood disorder.
Mood disorders may include persistent sadness or depression. Individuals may also experience loss of interest in daily activities. Changes in sleep patterns and appetite are common. Fatigue and difficulty concentrating may also occur.

20. Describe two differences between jet lag and travel fatigue.
Jet lag is caused by disruption of the circadian rhythm due to crossing time zones. Travel fatigue results from physical exhaustion during travel. Jet lag affects sleep patterns and hormone cycles, while travel fatigue mainly causes tiredness. Jet lag takes longer to recover compared to travel fatigue.

21. What is infradian rhythm?
Infradian rhythm refers to biological rhythms with cycles longer than 24 hours. These rhythms occur less frequently than once a day. An example is the menstrual cycle in females. They are regulated by internal biological clocks.

22. What is meant by biological rhythm in living organisms?
Biological rhythms are regular cycles of physiological processes in living organisms. They are controlled by internal clocks and environmental cues. These rhythms include circadian, infradian, and ultradian rhythms. They help organisms adapt to daily and seasonal changes.

Part C Short Essay Questions Answer any 5 questions Each question carries 4 marks

23. Write a short note on environmental zeitgebers.

Environmental zeitgebers are external cues that help regulate and synchronize the internal biological clock with the external environment. The most important zeitgeber is light, which influences the circadian rhythm by acting on the suprachiasmatic nucleus of the brain. Other zeitgebers include temperature, feeding schedules, physical activity, and social interactions. These cues ensure that physiological processes such as sleep, hormone secretion, and metabolism occur at appropriate times. Disruption of these zeitgebers, such as during shift work or long-distance travel, can lead to circadian rhythm disorders like insomnia and jet lag.

24. Explain the role of melatonin in sleep–wake regulation.

Melatonin is a hormone produced by the pineal gland that plays a crucial role in regulating the sleep–wake cycle. Its secretion is controlled by the suprachiasmatic nucleus and is stimulated by darkness while inhibited by light. Melatonin promotes sleep by lowering body temperature and reducing alertness, thereby preparing the body for rest. It also helps in maintaining the timing and quality of sleep. Any disruption in melatonin secretion, such as exposure to artificial light at night, can disturb sleep patterns and lead to sleep-related disorders.

25. Write on natural active immunity.

Natural active immunity is acquired when an individual is naturally exposed to a pathogen, leading to the activation of the immune system. The body produces specific antibodies and memory cells in response to the infection. This type of immunity is usually long-lasting and may even provide lifelong protection against certain diseases. For example, recovery from

infections like chickenpox or measles results in natural active immunity. It is an important mechanism by which the body defends itself and builds resistance to future infections.

26. What is Type II hypersensitivity and explain antibody-mediated cytotoxicity in Type II hypersensitivity.

Type II hypersensitivity is an immune reaction in which antibodies are directed against antigens present on the surface of cells or tissues, leading to their destruction. It is also known as cytotoxic hypersensitivity and involves IgG and IgM antibodies. In antibody-mediated cytotoxicity, these antibodies bind to target cells and activate the complement system, resulting in cell lysis. Additionally, immune cells such as natural killer cells may destroy the antibody-coated cells through antibody-dependent cellular cytotoxicity (ADCC). This mechanism is seen in conditions like hemolytic anemia and mismatched blood transfusion reactions.

27. Define secondary sexual characteristics. Enumerate the secondary sexual characteristics in females.

Secondary sexual characteristics are physical features that develop during puberty under the influence of sex hormones and distinguish the two sexes, though they are not directly involved in reproduction. In females, estrogen plays a major role in the development of these characteristics. These include enlargement of breasts, widening of hips, and increased deposition of subcutaneous fat, especially in the thighs and buttocks. Growth of pubic and axillary hair and the onset of menstruation are also important features. These changes indicate sexual maturity and prepare the body for reproductive functions.

28. Describe female fetal sex differentiation with reference to genetic makeup and hormonal environment.

Female fetal sex differentiation is primarily determined by genetic and hormonal factors. In the absence of a Y chromosome and the SRY gene, the undifferentiated gonads develop into ovaries. Since there is no secretion of testosterone or anti-Müllerian hormone, the Müllerian ducts persist and develop into female reproductive structures such as the uterus, fallopian tubes, and upper vagina. The Wolffian ducts regress due to the lack of male hormones. Thus, the default developmental pathway leads to female differentiation when male-specific hormonal influences are absent.

29. Enumerate the principal features of adaptive immunity and explain their significance.

Adaptive immunity is a highly specialized defense mechanism characterized by specificity, memory, diversity, and self/non-self recognition. It involves lymphocytes, particularly B cells and T cells, which respond specifically to particular antigens. One of its key features is immunological memory, which allows the body to respond more rapidly and effectively upon subsequent exposure to the same antigen. It also has the ability to distinguish between the body's own cells and foreign invaders, preventing autoimmune reactions. These features make adaptive immunity highly efficient in providing long-term protection and are the basis for vaccination.

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